



SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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GREEN COMPUTING & SUSTAINABLE IT

CSA 1007 SOFTWARE ENGINEERING

CO5 ASSESSMENT TOOL 2

GUIDED BY

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What is Green Computing?

Green Computing is the practice of designing, building, using, and disposing of computers and IT systems in ways that minimize environmental impact and maximize energy efficiency.

- **Energy-Efficient Hardware**

Designing devices and data centers that consume less power per task.

- **Sustainable Software**

Writing code and running processes that reduce computational waste.

- **Responsible E-Waste Management**

Recycling and safely disposing of outdated electronics.

- **Virtualization & Cloud Efficiency**

Consolidating workloads to cut hardware and energy needs.

Key Strategies for Sustainable IT

Four pillars that reduce environmental impact across the technology lifecycle



Power Management

Use energy-efficient processors, enable sleep modes, and adopt renewable power sources for data centers.



Data Center Optimization

Apply virtualization, efficient cooling, and workload consolidation to cut server energy use.



Sustainable Hardware

Choose durable, upgradable devices and extend product lifespans instead of frequent replacement.



Responsible E-Waste Disposal

Recycle components safely and partner with certified e-waste recovery programs.

Why It Matters: The Benefits

Sustainable IT delivers measurable gains for organizations and the planet

30%+

Lower energy costs

achievable through virtualization and efficient cooling in data centers.

50%

Longer device life

when hardware is properly maintained, upgraded, and reused.

Millions of tons

E-waste reduced

each year through responsible recycling and disposal programs.

Lower

Carbon footprint

for businesses adopting cloud and renewable-powered infrastructure.



The Path Forward

Conclusion & Next Steps

Green Computing is no longer optional — it's essential for reducing costs, meeting regulations, and protecting the environment. Organizations that act now gain both efficiency and reputation.

- 1 Audit current energy use and hardware lifecycle across your IT infrastructure
- 2 Migrate suitable workloads to efficient, renewable-powered cloud services
- 3 Set clear e-waste recycling and sustainable procurement policies
- 4 Track and report progress using recognized sustainability metrics

Think Green. Compute Smart. Sustain the Future.